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Incremental and decremental active learning for optimized self-adaptive calibration in viscose production

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ABSTRACT

In viscose production, it is important to monitor three process parameters as part of the spin bath in order to assure a high quality of the final product: the concentrations of H_2SO_4, Na_2SO_4 and $ZnSO_4$. NIR-spectroscopy is a fast analytical method applicable to conditions of industrial production and is capable of determining those concentrations. The collective composition of the spin bath varies in the industrial process, which implies changes in the matrix of the aforementioned analytes. Thus, conventional static chemometric models, which are trained based on collected calibration spectra from Fourier transform near infrared (FT-NIR) measurements, show a quite imprecise behavior when predicting the concentrations of new on-line data. In this paper, we are presenting a methodology which is able to cope with on-line self-calibration and -adaptation demands in order to compensate high system dynamics, reflected in conceptual changes in the mappings between NIR spectra and target concentrations. The methodology includes intelligent strategies for actively selecting those samples which should be accumulated into and excluded from the current data window in order to optimize the generalization performance of calibration models (thus termed as *incremental and decremental active learning stages*) while keeping the number of update cycles (and thus required target measurements) as low as possible. This follows the company requirements in terms of necessary *cost reduction*. Experiments on real-world data streams from viscose production process show that the new self-calibration methods are able to significantly reduce the number of update cycles while still keeping the predictive quality of the calibration models high (below 5% errors) for H_2SO_4 and Na_2SO_4. Incremental active learning is able to smoothen and improve the overall quality of the predictions, while decremental active learning achieves a lower number of medium to large prediction errors.

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1. Introduction

1.1. Motivation and problem statement

The viscose process is of economic significance and production has been growing rapidly for the past two decades. In the viscose production there is a demand for measuring the most important process parameters, namely the concentrations of H_2SO_4, Na_2SO_4 and $ZnSO_4$. The acid and the two salts govern the precipitation and agglomeration of the cellulose from viscose solution and the formation of the viscose fiber. The concentration of those components has major influence on the fiber properties. Hence, the accurate knowledge and control of the concentration are a prerequisite for the production of high-quality viscose fibers in the industrial processes. The conventional method to determine those concentrations is titration [1], which is quite time consuming, although it can be implemented in an automated fashion, i.e. automated titration draws a liquid sample from the process line and accomplishes an analysis on a regular basis. This is the case in the process under consideration in this work. The automated titration delivers an analytical value each 8th minute, whereas NIR-spectroscopy is capable to deliver a spectrum and consequently an analytical result each 10th second. The spin bath in the spinning trough is recirculated approximately each 20 min, compared to which the analysis time of the conventional titromat is quite long, what results in long reaction times to adjust the spin bath composition. This reaction time for adjustment could be dramatically shortened by the fast analysis achievable by

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NIR-spectroscopy. With the usage of chemometric models based on NIR spectra, a more frequent supervision of the process is possible. Furthermore, the automated titration also entails considerable costs for chemicals and maintenance, thus the reduction of its usage by automatic quantification based on chemometric models is highly desired. A complete substitution of the automated titration would be the ideal situation, however due to the high system dynamics of the spin bath, self-adaptation of the chemometric models is expected to be required from time to time (which could be verified in a former publication [2]). This is mainly because operating modes and states of the industrial processes might vary according to different products (i.e. fiber types) or raw materials processed, hence also the overall composition of the spin bath in detail might vary. E.g. the spin bath might contain additives or degradation products of the cellulose to a varying content [3], what is also conditioned by the dynamics of the industrial continuous operation mode and the recirculation of process media. These fluctuations in spin bath composition represent changes of the matrix of the analytes H_2SO_4, Na_2SO_4 and $ZnSO_4$. For NIR-spectroscopy these fluctuations might represent a problem, as a NIR-spectrum contains comprehensive information about the chemical composition, and specific analytical information is extracted by chemometric models. Due to the high expense for performing titrations, a requirement of the company is to reduce measurements and thus the update cycles. Ideally, the goal would be only to measure and update the model 3–4 times a day, i.e. each 6th to 8th hour instead of each 8th minute. Summarizing, we should achieve a double requirement consisting of capturing the high dynamics of the system, as well as reducing costs by decreasing the frequency of the titrations.

1.2. State-of-the-art and open problems

Ideally, when it comes to cost reduction, the best option would be to apply batch models. These are usually trained once based on off-line calibration samples, which are supposed to be representative of the whole process under study, and then applied to new on-line data for quantification and prediction of substances. Unfortunately, the dynamics of these matrix effects abandons the application of standard chemometric modeling tools, as e.g. can be found in [4,5] or [6] and have been applied to viscose production in the past [7,8]. In fact, in [2], it could be verified that, when using conventional modeling methods such as PLS [9], iPLS [10], PCR [11], LWR [12] and others (most of them used as implementations in the PLS-Toolbox[1]) in the viscose production process under our study, the model errors start to significantly drift over time. For the first few hundred of new on-line samples, the performance was similar to the one obtained on off-line training samples, but the more deteriorated the more time passed by (→ drift effect). Thus, focusing on capturing the dynamics of the process, in [2] a new concept was developed, termed as *evolving chemometric models* (*eChemo*) which possess the ability to automatically self-adapt and re-calibrate based on newly recorded on-line NIR spectra samples with minimal resources. The main characteristics of *eChemo* were:

- It required an initial model (a non-linear Takagi-Sugeno fuzzy system [13]) from some pre-recorded and off-line stored calibration samples. The parameters of the initial models (especially structure, i.e. number of rules, and input dimensionality) were optimized using a best-parameter grid search scenario coupled with cross-validation and specific model selection scheme, see [14].
- It was able to adapt the final selected model with new incoming on-line samples in a single-pass incremental manner, based on target concentrations measured by the automated titration. An interesting finding was that the model updates require evolution (and back-pruning) of structural components on demand [15] as well as forgetting of older samples [16,17] in order to increase both, compactness [18] and flexibility (condemning the usage of recursive linear models as used before in [19] and [20]). This assured that the model was always freshly up-to-date during the on-line process and a fast learning could be achieved.
- It performed an update with each new incoming sample (approx. each 8th minute), thus requiring permanent values from the automated titration. The quality of the predictions remained on a very high level (max. 3% relative error) during the whole process, thus prevented any drift effects. Regarding cost reduction, as can be seen in [2], when testing *eChemo* on updating each 8th or each 16th sample (far even from the company requirements of adapting 3–4 times a day, i.e. each 60th–45th sample) the performance loss is huge.

Therefore, neither the State-of-the-art batch models nor *eChemo* could find an acceptable trade-off between accuracy and cost. The former because of low accuracy due to the drift effect, and the latter because of an unacceptably high cost within a dynamically changing environment. Thus, new approaches should be investigated.

1.3. Our approach

In this paper, we demonstrate a different approach which is based on self-adaptive calibration models within a sliding window concept (Section 3.1), which puts more flexibility in the incremental learning phase and especially also in the forgetting process of older samples. We include more control on the issue which samples to forget than in the usual forgetting strategies, only based on the age concept (timeline). This is achieved by a *decremental active learning stage* (Section 4.2) which selects that sample from the window with lowest information gain in terms of diversity and model quality. Furthermore, the sliding window concept allows the introduction of more flexibility in terms of self-adjusting learning parameters and self-optimizing input dimensionality for the self-adaptive calibration models, thus allowing the integration of an input structure change, which is not possible in batch modeling nor in *eChemo* (in which the input structure is optimized in an initial off-line calibration phase, and then is kept fixed for the on-line phase). Finally, in order to put control on the samples requested for the self-adaptive calibration models and to avoid unnecessary updates not bringing any performance (e.g. samples already included in the model updates), we pursue a strategy which we term as *incremental active learning stage*. Therefore, we do not use each equidistant sample to be included in the sliding window, but we actively select samples with the highest information gain compared to the samples already present in the current window, i.e. samples which are expected to be able to generalize the chemometric models and thus to improve their predictive quality on new query points most. The number of actively selected samples has to be constrained in order to meet the requirements by the company, namely to record maximally 3 to 4 target values per day (thus allowing maximal 3 to 4 updates of the sliding window per day). As model architectures, we will exploit conventional PLS method (from the PLS-toolbox) and fuzzy systems with the batch model variant FLEXFIS as learning engine (as used in [2]), but employing latent variables from PLS as inputs. Thus, it is termed as FLEXFIS + PLS which can be seen as a sort of non-linear version of PLS; this will provide us with the information whether non-linearity in the self-adaptive calibration models helps to improve the performance.

The results include a comparison of the new sliding window approach with the original *eChemo* models [2], with different variants for deleting samples from the current windows (see Section 6), namely oldest versus random versus our enhanced selection strategy within the scope of *decremental active learning* as well as with different variants for accumulating samples in the window, namely blind static equidistant (default sliding window approach) versus the adaptation of an approach in [21] (based on Euclidean distances) versus our enhanced

[1] http://www.eigenvector.com/software/pls_toolbox.htm.

dynamic selection strategy within the scope of *incremental active learning*—note that only the sample selection strategy proposed in [21] is adapted to single-pass criterion (thresholding the distance), as the complete approach in [21] cannot be directly used as iterating over a pool of reference samples a multiple times (not available in our fully on-line single-pass setting). Several other active learning approaches proposed in literature such as e.g. [22–24] are also pool-based approaches, often iterating over the reference data base a multiple times, thus also not applicable for our purposes (on-line learning and processing).

This will be achieved based on 6 data streams, on-line recorded and reflecting the on-line viscose production process over a time horizon of about 6 months. The more flexibility of the sliding window based approach over *eChemo* pays off in significantly lower model errors, especially when trained only on each 6th–8th hour, i.e. 3 to 4 times per day (as requested by the company). Based on statistical significance tests, it could be also verified that the *decremental active learning* strategy is not significantly better than when leaving always the oldest out in terms of the average percentual prediction error over the whole stream. However, based on the error histograms it could be verified that large errors are less frequent when using our active selection. Another important result is the smoothing factor of predictions when using the *incremental active learning* strategy: high prediction errors which happen in case of blind equidistant sample accumulation and which cause peaks in the prediction curves over time disappear completely.

2. The setup

2.1. Description of the process

In the viscose process, pulp, which basically consists of cellulose, is processed to viscose fiber, which also consists of cellulose. Cellulose decomposes before melting. Therefore, forming of cellulose has to be realized by dissolution of cellulose and a regeneration. Hence the viscose process is a transformation of cellulose to the fiber form. The viscose fiber may be processed to various products of daily life, for instance clothes, but also to sanitary products (e.g. tampons) or textiles for the automotive industry. In the first step of viscose production cellulose or pulp, respectively, is steeped with caustic lye, in order to activate the cellulose for further chemical reaction by deprotonation. After pressing of the caustic cellulose slurry, the so-called alkali cellulose is yielded, which is stored for several hours in order to adjust the molar mass of the macromolecule cellulose for viscose production. The matured alkali cellulose is then treated with carbon disulfide, whereby the cellulose reacts with carbon disulfide to form xanthogenate groups. These xanthogenate groups enable the dissolving of the viscose in caustic lye, what happens in the next step, in which the so-called viscose is formed, a viscose solution of xanthogenated cellulose in caustic lye. The actual viscose fiber is formed by spraying the viscose via the so-called spinneret into the spinning trough, which is filled with the spin bath. The main constituents of the spin bath are sulphuric acid (H_2SO_4), sodium sulfate (Na_2SO_4) and zinc sulfate ($ZnSO_4$). The acid induces the splitting off of the xanthogenate group or carbon disulfide, respectively, from the cellulose, whereupon the cellulose precipitates and the viscose fiber is formed. Sodium and zinc sulfate in principle govern the speed, with which the viscose fiber is precipitated, what is crucial for the fiber property. After the spinning step the viscose fiber is finished by various after treatment steps, as thorough washing and drying.

2.2. Data acquisition

To realize a measurement setup all components utilized for the process monitoring were installed directly in the chemical plant, which means that a real in-line application has been set up. The utilized immersion probe (Hellma) is fiber optically connected to the FT-NIR process spectrometer and to a light source. The probe has an optical path length of 1 mm, with a measurement window processed out of sapphire. The FT-NIR process spectrometer [25] is constructed on the bases of a Michelson interferometer. In order to obtain a robust design, a monolithic interferometer [26] is utilized in the Michelson setup. With this technology all optical components are fused with a Quartz base plate, which features a setup with minimal thermal fluctuations. Furthermore, it is fully equipped with different data interfaces to enable, e.g., a communication with the process automation systems or to facilitate remote access from the office. The implemented programmable logic controller handles time critical signals as well as the communication to, e.g., a network attached storage server (NAS). Especially for time evolving system, as considered in this contribution, it is important to have an accurate timely synchronization between the analytical reference values and measured IR-spectra. For this task a digital signal provided by the process automation system was read in via an automation clamp, which was connected to process spectrometer through the CANopen interface. Once the chemical analysis has been carried out the reference values related to a certain spectral measurement (time of sample withdrawal) are saved automatically and thus fully synchronized with the NIR spectra.

The FT-NIR process spectrometer, whose maximal immersion depth is 690 mm, the minimal one 30 mm, the outer diameter 25 mm and the pipe length of the probe 700 mm, was continuously operated in a wavelength range from 3900 cm^{-1} to 11,000 cm^{-1} with a spectral resolution of 3 cm^{-1}. The spectral region is defined by the extended InGaAs — infrared detector implemented in the system. The measurement rate was 10 measurements per second. The spectra were calculated from an averaged interferogram of 100 single shot interferograms (measurement time: 10 s). A schematical sketch of the setup is shown in Fig. 1.

3. Basic methodology

3.1. Self-adaptive calibration models with sliding windows

The self-adaptive calibration models with sliding window concepts that we are proposing in our paper do not require an initial off-line stage, thus can be used from scratch in a kind of plug-and-play manner, and also is able to dynamically adjust the optimal input structure of calibration models. Therefore, consider as input a data stream from the on-line process $(\vec{x}_1, y_1), (\vec{x}_2, y_2), \ldots$, with $\vec{x}_1$ the first data sample containing a full spectrum, i.e. a vector of wavelengths and y_1 the continuous target concentration value, thus being confronted with a data stream regression problem [27]. In principle, a data stream is unbounded in its size, arrives continuously and the order of samples is predetermined (see also [28] for a detailed definition). In our case, we are restricted to a sliding window W_0 containing d number of samples, thus

$$W = \{(\vec{x}_i, y_i) | i = 1, \ldots, d\} \tag{1}$$

If the sliding window is moved over a data stream in a classical way (without any methodological intervention), $(\vec{x}_1, y_1)$ denotes the oldest sample and $(\vec{x}_d, y_d)$ denotes the newest sample; d should be chosen depending on the model architecture used for calibrating: for non-linear models usually more samples are requested to obtain reliable models than for linear ones, see also [29]. In all our test experiments (see Section 5), we varied this value $d = 50, 100, 200$, also in order to observe the impact on the stability of non-linear models.

At the start, the window is initialized with the first d samples (spectra + target values measured by the automated titration) which we denote as W_0, which is filled with new incoming stream samples from time to time—in our case, according to the maximal frequency usage of the automated titration required by the company, i.e. each 6th to 8th hour = each 45th to 60th sample. Assuming to have a

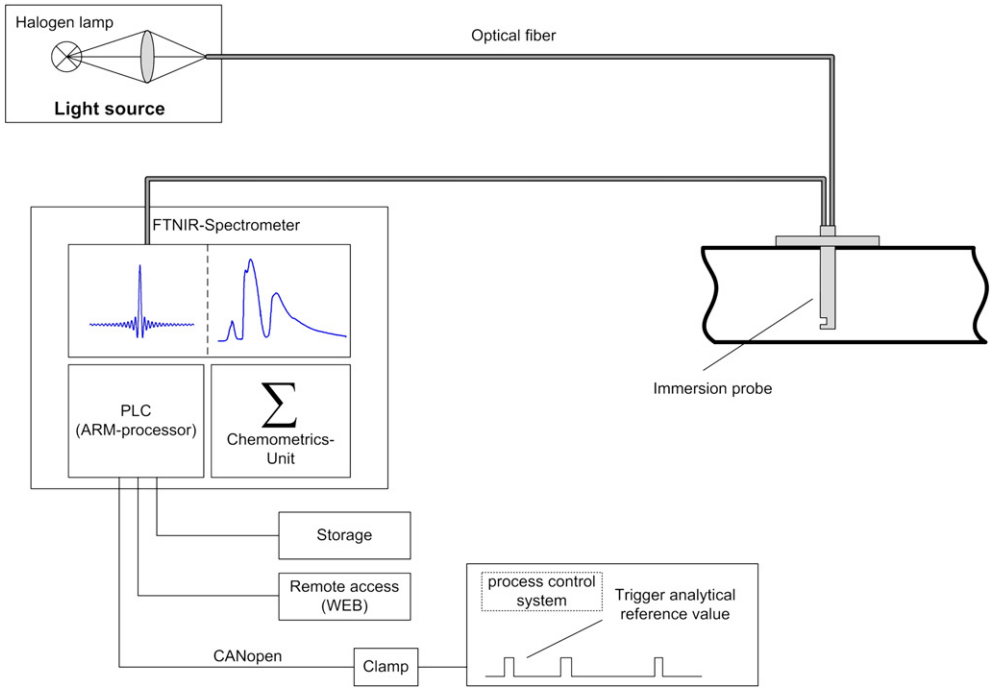


Fig. 1. Schematic view of the data acquisition framework as installed in-line in viscose production.

window W_t at time instance t established, our modeling procedure performs the following steps:

1. Calculate latent variables $lat_1, \ldots, lat_{all}$ with *all* the number of wavelengths contained in the spectra, and ordered according to their importance. Notice that our approach includes as regression algorithms both PLSR and FLEXFIS + PLS (see [30] for the basics of FLEXFIS). Therefore, the latent variables from PLS are always needed.
2. Define parameter grid: Parameter selection for PLSR and FLEXFIS + PLS is based on a grid search including a cross validation procedure. The parameter grids are different for each of the algorithms:
   - For PLSR, use $dim = \{1, \ldots, a\}$ for the number of latent variables to be included into the calibration model, the upper limit a is set to 30, achieving a vector of grid points $\vec{g}_i = dim_i$.
   - For fuzzy systems, use the number of latent variables and define the vigilance (parameter that controls the level of non-linearity

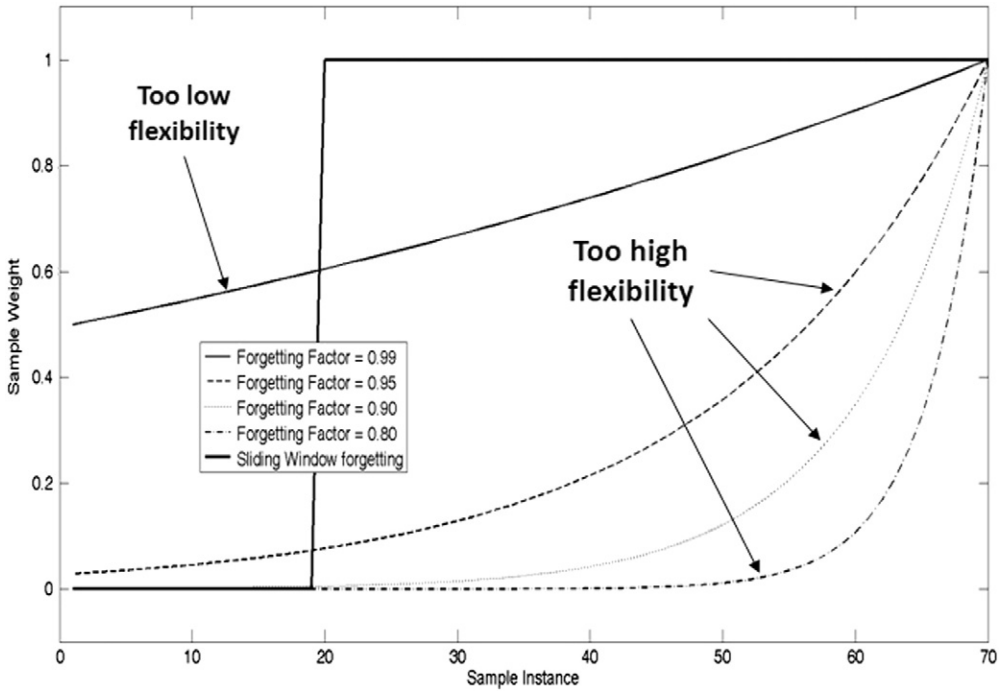


Fig. 2. Forgetting strategies achieving different weights for past samples: when using a forgetting factor value of 0.8–0.95 in *eChemo* approach, only the latest 10–20 samples are really affecting model learning (too much flexibility); when increasing it toward 1, older samples are also included into the model with high weight (too less flexibility).

applied [30]) $vigi = \{0.1, \ldots, 0.9\}$ for the learning engine, achieving a matrix of grid points $G_{ij} = (dim_i, vigi_j)$.

3. For all grid points, perform 10-fold cross-validation [31] and store the cross-validation error: $CVerr_i$ respectively $CVerr_{ij}$.
4. Perform model selection. Complexity increases with the number of latent variables in PLSR. In FLEXFIS + PLS, it increases in a direct way with dimensionality and in an inverse way with vigilance. Our model selection procedure selects the parameters corresponding to the grid point for which the corresponding model has lower CVRMSE, after being penalized according to their complexity ($CVerr(pen)$):

$$CVerr_{i(j)}(pen) = CVerr_{i(j)} \cdot e^{\alpha \quad param1+\beta \quad (1-param2)} \tag{2}$$

with *param1* the number of inputs and *param2* the vigilance in case of fuzzy systems; for PLSR, $param2 = 1$. α and β are normalization factors which are set to 0.05 in our case, 0.5 respectively. Thus, we obtain

$$\begin{aligned} dim* &= argmin_i(CVerr_i(pen))(dim*, vigi*) \\ &= argmin_{i,j}\left(CVerr_{ij}(pen)\right) \end{aligned} \tag{3}$$

5. Perform a final model training with $dim*$ for PLSR resp. ($dim*$, $vigi*$) for fuzzy systems on all d samples from $W_t \rightarrow$ new calibration model f for predicting concentrations for newly recorded spectra (permanent supervision).
6. Determine the sample that will be introduced in the next window (ingoing sample) by means of the *incremental active learning stage*, in Section 4.1.
7. Once the ingoing sample is known, determine the sample in the current window that is going to be replaced by it (outgoing sample) by means of the *decremental active learning stage*, in Section 4.2.

Depending on the applied model architecture, the calibration steps (within cross-validation and in the final training cycle) differ. In the case of non-linear fuzzy systems, we apply the FLEXFIS batch modeling engine [32] for chemometric purposes, i.e. using the ordered latent variables (most important first) obtained by PLS as inputs, thus termed as FLEXFIS + PLS. In particular, each sample from W in Eq. (1) is projected to $\vec{x}_{proj}$ by applying

$$Lat_a = [\vec{lat}_1 | \cdots | \vec{lat}_a] = X * V_{X,a} \Rightarrow \vec{x}_{proj} = \vec{x} * V_{X,a} \tag{4}$$

The projected samples (scores) are sent into the FLEXFIS learning engine, with a the number of latent variables in the currently parameter grid search step, and V the loadings of the wavelengths in each latent variable. This neglects the usage of single wavelengths (as done in our previous chemometric calibration settings [2,14]), which may be biased due to noise. As d is small, the computational demand is limited, and thus the computational problems when conducting re-training on past samples (as discussed in [2]) do not arise. Furthermore, a full parameter grid search always guarantees optimal learning parameters and model structures (input dimensionality + number of rules in case of fuzzy systems) at each stage of the process. For further details on the FLEXFIS learning engine with all its extension, please refer to [32]; for details

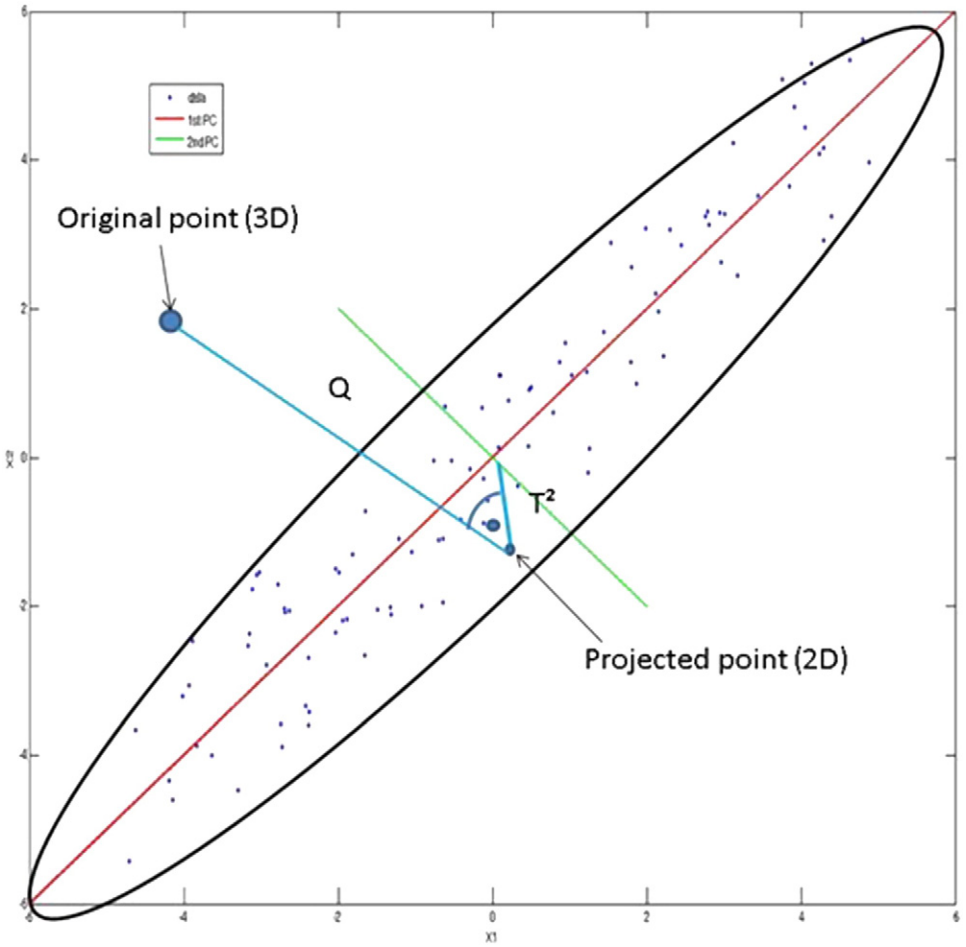


Fig. 3. Visualization of the meaning of Q-statistics and T^2-statistics for a three-dimensional example projected onto a two-dimensional subspace.

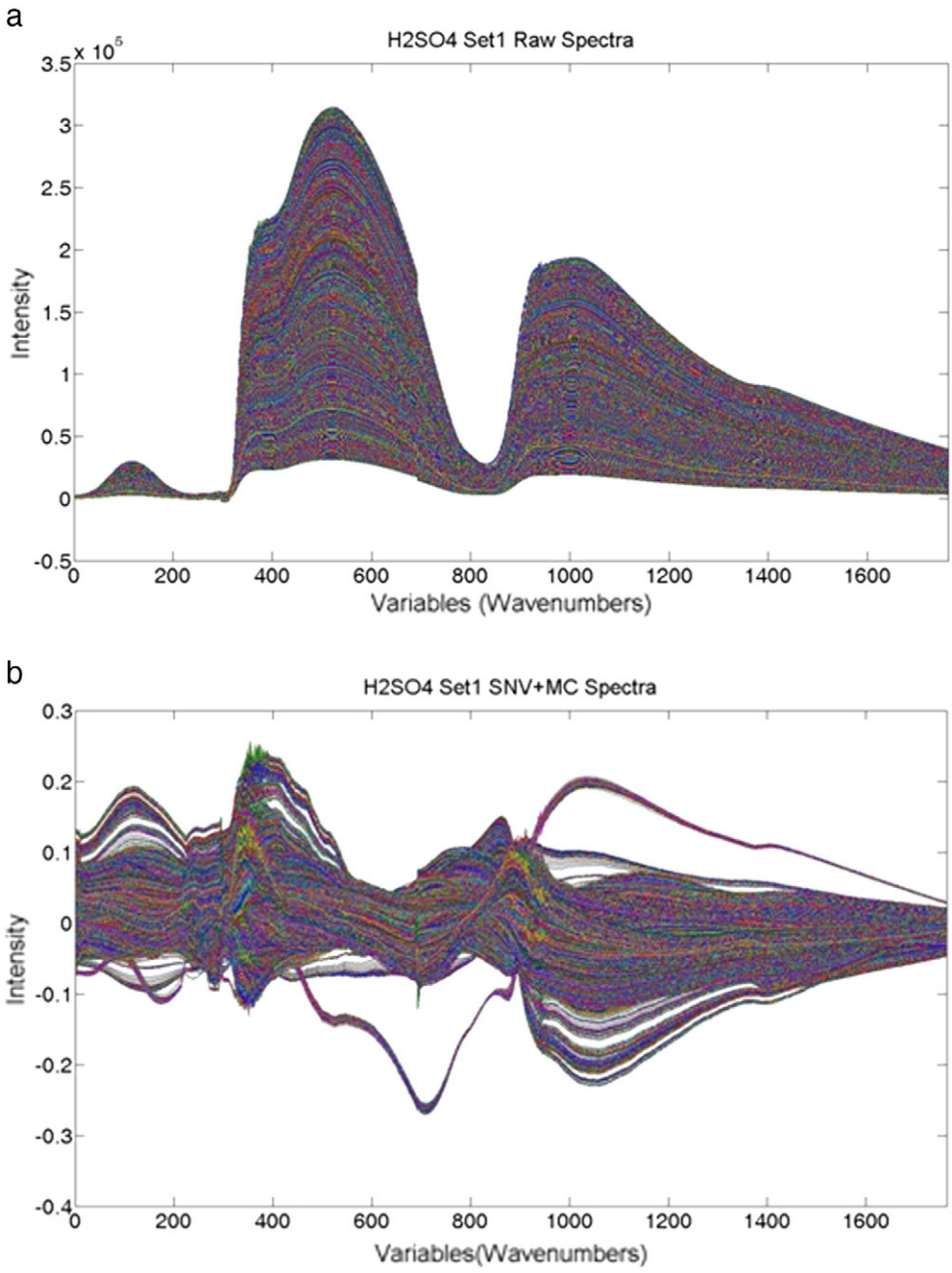


Fig. 4. Original spectral samples (a) and pre-processed ones according to SNV and mean centering (b) for H_2SO_4.

on its concrete application to chemometric modeling from FT-NIR spectra, please refer to [14] and [33].

The windowing approach can be also interpreted as a kind of forgetting strategy, where the last *d* samples are receiving a full weight of 1 in the learning process and all the others a weight of 0, thus are completely forgotten. Finally, the much better performance of this forgetting strategy over the usual exponential forgetting (employed in *eChemo*) served as basic motivation for its on-line implementation, verification and description in this paper. We see a reason for this in the fixed forgetting factor used in the exponential forgetting strategy: i) setting the forgetting factor parameter to a too low value leads to (too) low weights for samples in the recent past (only 4,5 time instances back → too much flexibility), ii) setting it to a value close to 1 includes too many past samples with a significant weight (loss of flexibility), such that the dynamics of the process cannot be sufficiently compensated. Fig. 2 shows this problematic. The sliding window approach connected with the active learning methodologies (as will be presented in the subsequent sections) offers the possibility to react more interactively on the actual process behavior, i.e. to accumulate those samples in the window (to be used for re-training) which in fact bring in additional information resp. some sort of diversity (e.g., caused by a change in the process), compared to the samples already in the window. In this sense, the sliding window may contain holes, i.e. some newer samples are not included, but on the other hand it still may carry important information which appeared more than *d* samples ago. Thus, we may also speak about an *active sliding window* approach.

On the other hand, due to the re-training steps (Steps 1 to 5 above), the sliding window approach is expected to be slower than single-pass sample-wise model adaptation. In Section 6 the computational burden of using re-training steps on sliding windows (instead of incremental

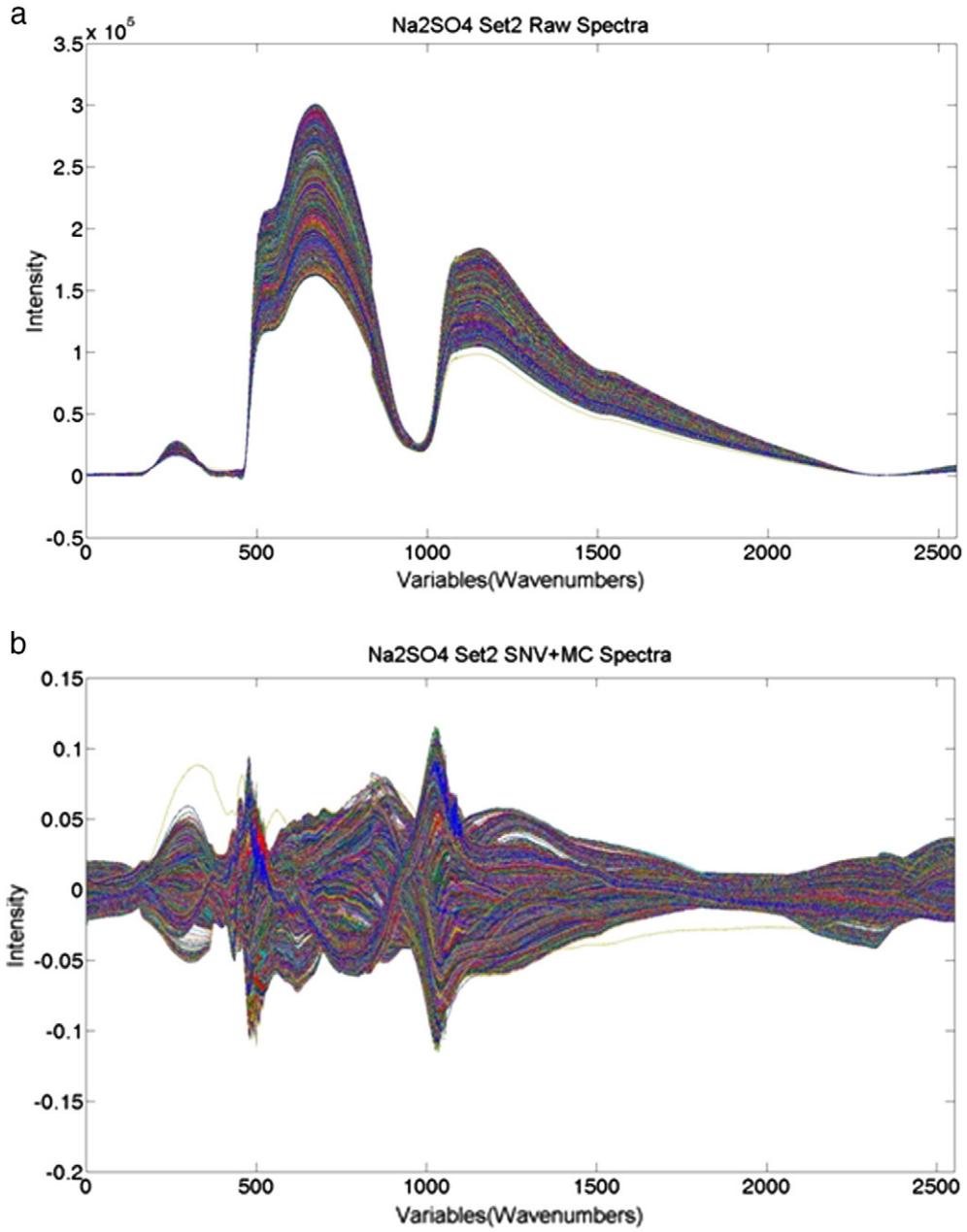


Fig. 5. Original spectral samples (a) and pre-processed ones according to SNV and mean centering (b) for Na_2SO_4.

stages) will be examined, also in connection with the minimal speed requirements from the company.

4. Single-pass active learning concepts for improved performance

The motivation for improved performance is to control which samples to accumulate in the current window and which samples to delete from the current window, instead of using blindly each *X*th sample for accumulation (with *X* specified by company requirements) and always the oldest samples in the window for deletion. A specific requirement in the on-line sliding window mode is that the selection strategies have to operate in a *single-pass manner*, which means that, based on a new incoming sample, a decision has to be made whether to include it and if so, which sample to delete in order to always keep *d* samples in the window and thus to be able to control the speed for the model re-training steps (growing windows would slow down significantly over time). The single-pass requirement is a strict requirement, as it does not allow to use any candidate spectra samples from the past (and then to select the best ones out of these). This single-pass approach is unavoidable because the cost optimization goal requires measuring as less target values (titration) as possible. Moreover, the target value of a past sample cannot be reproduced, as it reflect a past spin-bath behavior and since then the process permanently continued its operation, thus measuring the target in the present in order to match it to an old spectrum sample is not an adequate option. In this sense, conventional active learning strategies as can be found in literature, for instance see [22–24,34–36], are not applicable as they typically iterate over a pool of samples to select the most informative ones (including re-training steps in batch off-line mode). Approaches such as [37,38] operate in on-line mode, the first one also requires a pool of past samples to select samples from time to time, the latter relies on reliability concepts based on evolving classifiers (without windowing), thus not being applicable in a regression context with windowing. Others are not unsupervised, meaning that the target values are used, like [39–41]. This

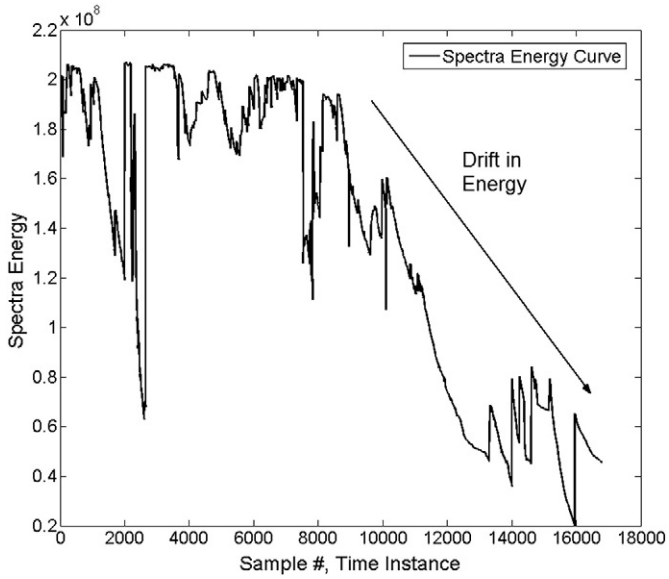


Fig. 6. Drift of the spectra energy over time in case of H_2SO_4 and the first portion of the data set (containing 16,799 samples).

is not compatible with the cost optimization goal. Furthermore, all of these are not operating in decremental stages, i.e. selecting samples for exclusion from learning.

4.1. Incremental active learning stage

The incremental active learning stage takes control on the samples which should be included in the sliding window in order to get best diversity for the modeling process. Thus, the goal is to introduce more intelligence than a blind equidistant accumulation of samples (yielding a classical sliding window approach). Of course, it has to be an unsupervised criterion as only in case when a sample is selected for accumulation, the automated titration is triggered to measure the corresponding target concentrations (H_2SO_4, Na_2SO_4 and $ZnSO_4$) for the current process state (to optimize costs for titration values).

The information gain a new sample adds to the already existing ones within the current sliding window should play a central role in the selection process. We employ the Q-statistics and Hotelling's T^2-statistics [42] from PCA, where Q represents the distance from the original point to the principal components projection subspace, and T^2 the distance from the projected sample to the center of the distribution inside the projection subspace determined by the first a principal components, spanned by

$$X = T_a P_a' + E_a \tag{5}$$

with T_a the scores matrix and P the loadings matrix for the a first principal components $P_a' = [p_1, p_2, \ldots, p_a]$ with $a < M$, M the dimensionality of the original data matrix, i.e. the number of wavelengths. For the subspace P_a' we use the same dimensionality a as applied in the current final calibration model extracted from the sliding window (elicited through best parameter grid search, see Steps 1 to 5 in Section 3.1). The Q-statistics is formally defined as:

$$Q = \vec{x} - \vec{x} P_a P_a^T \tag{6}$$

with P containing the loadings of the a principal components. The Hotelling's T^2-statistics is formally defined as:

$$T^2 = \sum_{i=1}^{a} \left(\frac{t_i}{\lambda_i}\right)^2 \tag{7}$$

with t_i the score of the ith component in current data sample and λ_i the eigenvalue of the ith component. Fig. 3 visualizes the meaning of both statistics for an original three-dimensional example projected onto a two-dimensional rotated space.

Now, if both, T^2 and Q-statistics are small for a new incoming sample $\vec{x}$, the sample falls close to the center of the projected sliding window samples. In this sense, $\vec{x}$ can be expected to be represented well by already existing samples and there is no need to update the window. On the other hand, if one of the two statistics is high, $\vec{x}$ denotes an important add-on to the current window as reflecting a new information (outside the old one); this could be, for instance, due to a dynamic system change or a new arising operation mode/cycle. For both statistical-funded distances, thresholds in the form of significance levels can be derived [43], namely

$$T_\alpha^2 = \frac{a(d-1)}{d-a} F_\alpha(a, d-a) \tag{8}$$

for the T^2 statistics, with d the window size and

$$Q_\alpha = \Phi_1 \left[\frac{c_\alpha \sqrt{2\Phi_2 h_0^2}}{\Phi_1} + 1 + \frac{\Phi_2 h_0 (h_0 - 1)}{\Phi_1^2}\right]^{1/h_0} \tag{9}$$

for the Q-statistics, with $\Phi_i = \sum_{j=a+1}^{M} \lambda_j^i$, $h_0 = 1 - \frac{2\Phi_1\Phi_3}{3\Phi_2^2}$ and c_α the normal deviate corresponding to the upper $1-\alpha$ quantile. We used a tolerant value of $\alpha = 0.99$ in all test experiments.

The final decision whether a potential candidate sample $\vec{x}_k$ with a high Q or a high T^2 is accumulated depends on the restriction deployed by the company for minimizing the costs for titration measurement values. Measurements should be taken maximally 3 to 4 times a day which is equal to an allowed re-train ratio (RR) between 45 to 60 samples, as each 8th minute a new sample is recorded, i.e. in average we are allowed to retrain every 45–60 samples. We could force the algorithm to wait until 45 samples appear before beginning to look for a new ingoing sample, leading to a range [45–60], but this would be too restrictive. We relax the lower boundary by using a range [$floor(60/c)$, 60], where $floor(x)$ is the first integer smaller or equal to x. By changing a, we can steer the trade-off between model updates and measurement costs (default in the experimental section will be $a = 3$, theoretically allowing a re-train ratio (RR) of 20, but we will see that in this case RR = 45 will be not underrun in all our test data stream cases). Thus, our condition for accumulating a sample in the window becomes:

$$\left(T^2 > T_\alpha^2 \vee Q > Q_\alpha\right) \wedge (floor(60/c) \leq k - k_{old} < 60) \tag{10}$$

with k_{old} the index of that sample which was last accumulated, and k the index of the current sample.

4.2. Decremental active learning stage

Once we have found a suitable sample $\vec{x}_k$ to be introduced in the window, a novel methodological content stems from the motivation to improve the sample selection regarding which samples to discard from the buffer. We have mentioned the option of simply discarding the oldest in the buffer, which is a plausible strategy, as especially in dynamic processes the oldest samples can often be assumed to be the least "informative" ones. However, it is not an optimal one, as for instance a new incoming sample could be very similar to another sample that is already in (not necessarily the oldest one). In this sense, the very similar sample can be discarded from the buffer as providing information for the model which is redundant to the new ingoing sample (thus, it would not improve the diversity of the model).

In order to verify the maximal quality gain of a model with respect to the exchange of $\vec{x}_k$ and one of the current calibration samples within the window W_t, we are aiming for a two-stage selection process:

1 Given $\vec{x}_k$, its similarity to all samples in the current window is calculated:

$$S_i = Sim(\vec{x}_k, \vec{x}_i) \forall i = 1, \ldots, d \tag{11}$$

with *Sim* a spectra similarity measure as derived below (see subsequent paragraph named *Spectral Similarity*). Then, it is checked whether the sample which has maximal similarity, i.e.

$$i* = argmax_{i=1,\ldots,d}(S_i) \tag{12}$$

is significantly similar to $\vec{x}_k$ in a statistical sense (by means of a hypothesis text, as explained below). If this is the case, $\vec{x}_{i*}$ is replaced by $\vec{x}_k$ in the window, thus

$$W_{t+1} = (W_t \backslash \{\vec{x}_{i*}\}) \cup \{\vec{x}_k\}. \tag{13}$$

2 If none of the samples in W_t is significantly similar to $\vec{x}_k$, then we use a so-called leave-one-sample-out (LOSO) strategy in order to check which sample is the less informative one, known that $\vec{x}_k$ will go in. We consider d subsets (as many as samples in W_t), each of them containing all samples bit one

$$\widetilde{W}^i_{t+1} = W_t \backslash \{\vec{x}_i\} \forall i = 1, \ldots, d \tag{14}$$

Since we know that $\vec{x}_k$ is going in, its target value will be provided by the automated titration, thus we can perform steps 1–5 in Section 3.1 for each subset $\widetilde{W}^i_{t+1}$ and obtain the residual of prediction Res_i. The subset performing better is the one in which the less informative sample is missing

$$i* = argmin_{i=1,\ldots,d} Res_i \tag{15}$$

So $i*$ is the index of the sample that should be replaced by $\vec{x}_k$ because the deletion of that sample has the smallest (increasing) effect on the model error over all samples.

4.2.1. Spectral similarity [44]

Measuring the similarity of two spectra is done based on hypothesis testing including two hypotheses pointing to whether the spectra are equal or non-equal. Formally, let s_1 and s_2 therefore be two high-dimensional spectra samples (including a wavelengths). Then, we establish the hypothesis test:

$$\begin{cases} H_0: & \Delta s = |s_1 - s_2| = 0 \\ H_1: & \Delta s = |s_1 - s_2| \neq 0 \end{cases} \tag{16}$$

In the ith wavelength, the intensity follows $N(s_i, \sigma_i^2)$ the real intensity and standard deviation in this wavelength. Therefore, scaling the data to unit variance and assuming independence in the measurements, $\Delta s_{i;1,2} = |s_{i,1} - s_{i,2}|$ is normally distributed with $N(0, 2)$. Considering the mean of the difference over all wavelengths, i.e.

$$T = \frac{1}{a}\sum_{i=1}^{a} \Delta s_i = \frac{1}{a}\sum_{i=1}^{a} |s_{i,1} - s_{i,2}| \tag{17}$$

its expectation and variance are $E(T) = 0$ and $Var(T) = \frac{2}{a}$. In this sense, we re-formulate the hypothesis test in Eq. (16) to:

$$\begin{cases} H_0: & E(T) = 0 \\ H_1: & E(T) = 3\sqrt{\frac{2}{a}} \neq 0 \end{cases} \tag{18}$$

taking into account that $3\sigma = 3\sqrt{\frac{2}{a}}$ covers 99.7% of the confidence. The posteriori probabilities according to Bayes formula, are

$$\begin{aligned} P(H_0|T) &= \frac{P(T|H_0)P(H_0)}{P(T|H_0)P(H_0) + P(T|H_1)P(H_1)} \\ P(H_1|T) &= \frac{P(T|H_1)P(H_1)}{P(T|H_0)P(H_0) + P(T|H_1)P(H_1)} \end{aligned} \tag{19}$$

Thus, the posteriori odds ratio becomes:

$$POR = \frac{P(H_0|T)}{P(H_1|T)} = \Lambda \frac{P(T|H_0)}{P(T|H_1)} = \Lambda LR \tag{20}$$

where the last quotient is termed as the likelihood ratio and $\Lambda = \frac{P(H_0)}{P(H_1)}$. Without a priori information, we can see the incidence of the events $P(H_0)$ and $P(H_1)$ as equal, thus $\Lambda = 1$. Furthermore, it follows:

$$\begin{aligned} P(T|H_0) &\cong N(0, 2/a) \\ P(T|H_1) &\cong N\left(3\sqrt{2/a}, 2/a\right) \end{aligned} \tag{21}$$

In this sense, the posteriori odds ratio becomes:

$$POR(s_1, s_2) = \Lambda \frac{\sqrt{a/4\pi} e^{-(a/4)T^2}}{\sqrt{a/4\pi} e^{-(a/4)\left(T - 3\sqrt{2/a}\right)^2}} = \Lambda e^{-\frac{a}{4}\left(6T\sqrt{2/a} - \frac{18}{a}\right)} \tag{22}$$

which is then used as similarity measure between incoming sample $\vec{x}_k$ and all other samples in the current window W, thus $Sim(\vec{x}_k, \vec{x}_i) = POR(\vec{x}_k, \vec{x}_i)$. Obviously, two spectra samples can be treated as equal whenever $POR(\vec{x}_k, \vec{x}_i) > 1$, as the numerator in Eq. (22) represents the likelihood that hypothesis H_0 is true. If more than one sample induces $POR(\vec{x}_k, \vec{x}_i) > 1$, that one with the highest *POR*-value is deleted from the window.

5. Experimental setup

5.1. Data set characteristics and pre-treatment

Data sets recorded on-line from the spin bath during the viscose production process are shown in Fig. 4 for H_2SO_4, and in Fig. 5 for Na_2SO_4, respectively, (a) the original ones as recorded and (b) the spectra obtained after SNV (standard normal variate) and mean centering.

Interesting to see, that the amplitude range of the whole spectral range in the case of H_2SO_4 is pretty wide (note that this is the overlay of 16,799 samples, its recordings ranging over a period of about 93 days or approximately 3 months). Thus, it can be expected that the process is quite dynamic and may change its behavior over time. In order to pursue this assumption, we also observed the change of the spectral energy (sum of the amplitude values over all wavelengths) over time, its behavior shown in Fig. 6, where clearly a drifting effect over the second half of the data set can be examined. This does not reflect a faulty operation case (which would last over about 1 1/2 month in this case), but a dynamic process change in the viscose production (different spin-batch cycles). Hence, a self-adaptive methodology adapting quickly through the drift phase is highly desired, as static models (built on the first half of the data, for instance) would meet a lot of extrapolation cases (projected bands moving into a lower amplitude region) and thus low quality predictions for new query points sent into their inference during the drift phase. We also already verified such occurrences in [2] for older data streams from the same process recorded during 2011.

Fig. 7 visualizes the development of the target values y over time for all three concentrations—note that these are currently still permanently measured during production and should be substituted by our active learning approach in the near future; by having these real measured values available in the data sets is important, as we can evaluate the

achieved error rates of our approaches, see the subsequent section. No drifts in the targets can be observed, but high dynamic changes in the case of $ZnSO_4$, which also means that this will be the hardest concentration to model, as we will see below in the results section. The summary information about the used data sets is shown in Table 1. The cut into two pieces of the 6 months data stream was due to the appearance of some technical problems in the installation, which caused a period of several days without data recording. Thus, considering the data set as a whole timeline data set would lead to wrong and/or unfair conclusions.

5.2. Evaluation scheme

Regarding the evaluation of our proposed approach in terms active learning, we conduct the sliding window based approach over the two data streaming sets separately, leaving us to analyze six results for the

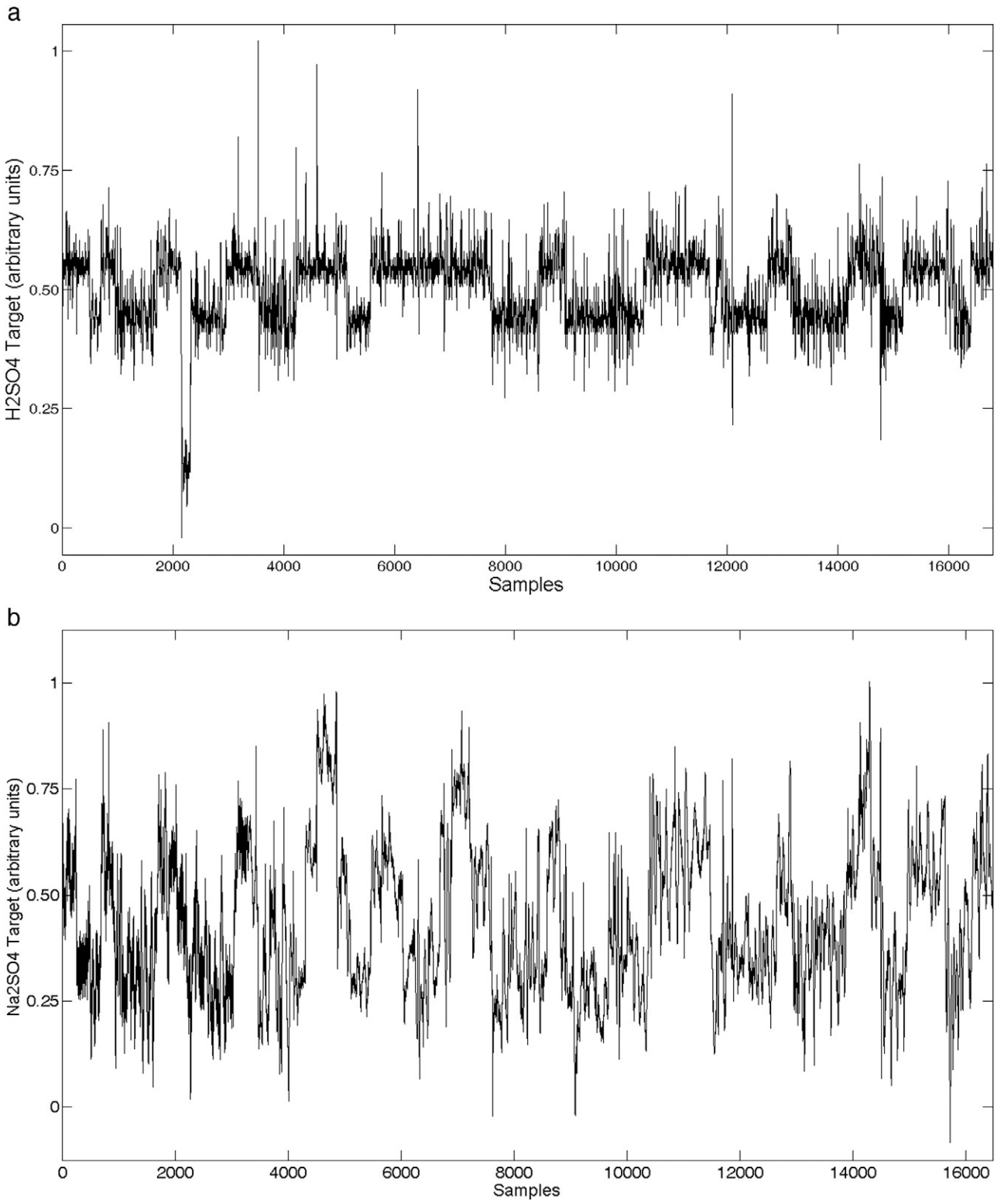


Fig. 7. Target values over time for H_2SO_4 (a), Na_2SO_4 (b) and $ZnSO_4$ (c).

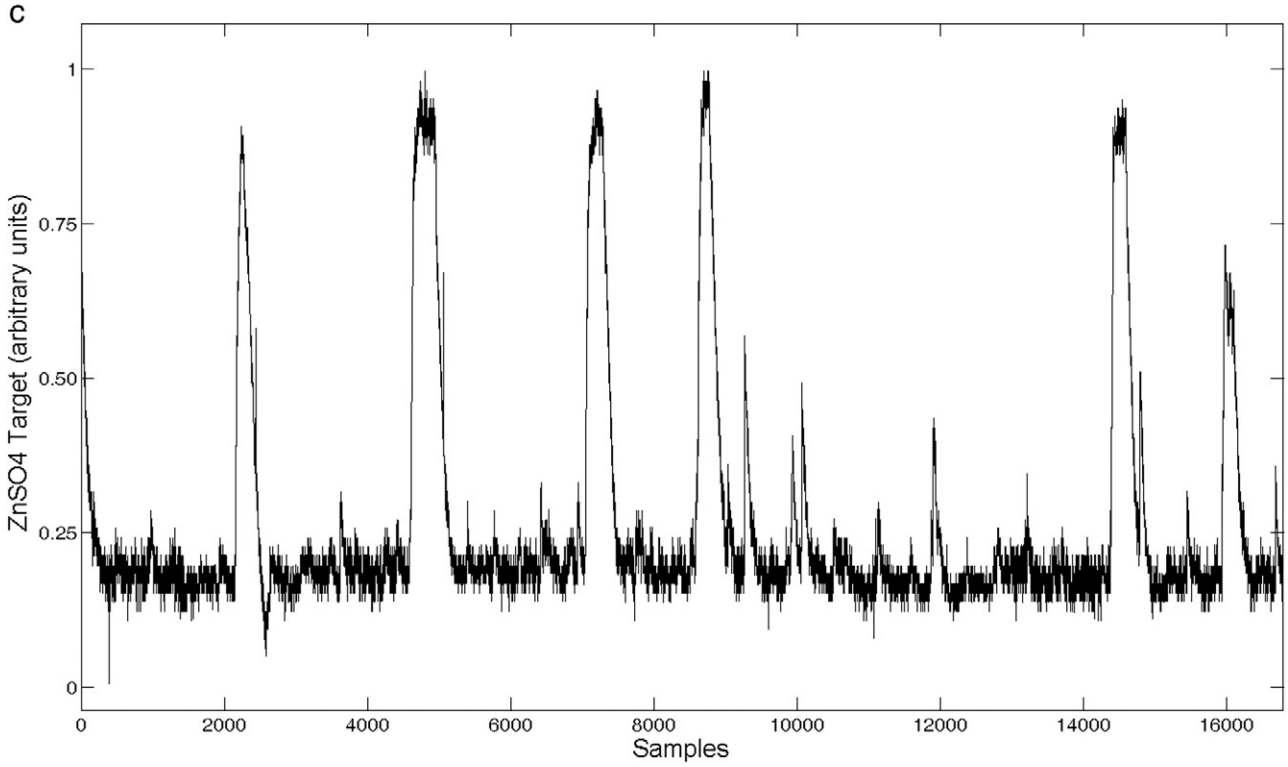


Fig. 7 (*continued*).

two streams and the three targets. In particular, the evaluation scheme on one data stream—target combination follows the following procedure, favoring a *self-adaptive learning from scratch* policy (no initial separate batch off-line modeling is needed):

1. Collect the first d data samples from the stream and store it into the window W defined by Eq. (1).
2. Conduct Steps 1 to 5 in the itemization of Section 3.1 to obtain the first calibration model.
3. For each new incoming data sample (spectrum) $\vec{x}_k$ at time instance k, do the following steps (Steps 4 to 7)
4. Obtain the prediction from the current calibration model $\hat{y}_k = f(\vec{x}_k)$ and store it and its residual to the real measured target value $Res_k = \hat{y}_k - y_k$; in this sense, we always obtain residuals on *separate validation data* (at least always RR = (60/c) steps ahead).
5. Check whether Condition (10) holds: if not, go to Step 3.
6. Accumulate $\vec{x}_k$ into the sliding window and discard sample with index according to Eq. (12) using Eq. (22) or Eq. (15) (if Eq. (22) is not fulfilled for any sample in W).
7. Perform re-calibration Steps 1 to 5 in the itemization of Section 3.1 to obtain an updated model f; go to Step 3

The above scheme is independent from the chosen model architecture, we will use PLSR and FLEXFIS + PLS as motivated in Section 3.1. The residual vector Res_k, $k = d+1$,...(obtained on separate future validation data) is then used for calculating overall error measures such as the normalized root mean squared error, i.e. the root mean squared error divided by the range of the target, histograms on the absolute error and performing statistical preference analysis.

5.3. Comparison with state-of-the-art

We will compare our method with the incremental learning approaches Incr-FLEXFIS + PLS (the most promising variant of *eChemo* paradigm), and incremental principal components regression (incrPCR) [45]. These approaches come with single-pass model updates including forgetting factors for achieving more flexibility in case of dynamic changes and drifts, and do not rely on a sliding window concept, thus no decremental learning stage is needed. They consist of an initial batch model which is subsequently updated. As the model is calculated only once, the amount of batch samples (BS) is not so relevant as in the window-based cases. Therefore, we fix it in $BS = 200$. In order to perform a fair comparison, we will use for both incremental non-window-based algorithms the same three ingoing sample selection strategies employed for the window-based ones

- *Static*: Using a fixed retrain ratio $RR = 60$.
- *Euclidean*: Based on [21], in which the authors determine which samples from a pool of candidates should be included in a training set according to the Euclidean distance from each candidate sample to the training set. As we cannot keep a pool of candidates, because of the single-pass condition, we must determine a threshold for the minimum Euclidean distance that determines the acceptance of the candidate sample as ingoing sample. We have defined the threshold as

$$Thres = \mu \cdot \max_{i=1,\ldots,d} (Dist(\vec{x}_i, C)) \tag{23}$$

where *Dist* is the Euclidean distance and C the center of the samples in the window (training set). The parameter μ is crucial in the frequency

Table 1
Summary data set characteristics.

	H_2SO_4 (Set 1)	H_2SO_4 (Set 2)	Na_2SO_4 (Set 1)	Na_2SO_4 (Set 2)	$ZnSO_4$ (Set 1)	$ZnSO_4$ (Set 2)
# Samples	16,799	9322	16,470	9150	16,798	9321
# Variables	1762	1762	2550	2550	2192	2192

Table 2
Normalized root mean squared errors (percentual errors) for PLSR (upper part) and FLEXFIS + PLS (lower part) in case of a window size of 50, using different variants for sample exclusion (oldest, random and our decremental active learning ("LessInfo")) and applying equidistant sample accumulation according to a re-train ratio of 60 ("Stat" columns), the approach in [21] ("Eucl" columns), and the exquisite selection according to our incremental active learning topology ("Dyn" columns). The values inside the parenthesis indicate the real averaged retrain ratio achieved (note that that value is fixed in 60 for "Stat"), and the values in bold font indicate violations of company requirement (max. 5% errors).

PLSR		Random			Oldest			LessInfo		
Target	Set	Stat	Eucl	Dyn	Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	1.35	1.35 (51)	1.32 (54)	1.34	1.33 (54)	1.31 (56)	1.34	1.36 (50)	1.25 (51)
H_2SO_4	2	1.14	1.16 (58)	1.11 (55)	1.11	1.09 (53)	1.05 (52)	1.06	1.05 (48)	0.95 (46)
Na_2SO_4	1	1.72	1.72 (59)	1.80 (56)	1.47	1.46 (49)	1.47 (54)	1.45	1.48 (54)	1.48 (55)
Na_2SO_4	2	1.68	1.66 (52)	1.66 (54)	1.71	1.69 (54)	1.52 (54)	1.66	1.52 (50)	1.47 (52)
$ZnSO_4$	1	**20.35**	**20.50** (46)	**20.53** (49)	**22.98**	**22.02** (52)	**21.53** (54)	**23.86**	**20.89** (49)	**20.88** (51)
$ZnSO_4$	2	**17.54**	**20.98** (47)	**22.46** (53)	**21.93**	**21.26** (50)	**21.13** (53)	**21.23**	**21.18** (54)	**21.05** (53)
FLEXFIS		Random			Oldest			LessInfo		
Target	Set	Stat	Eucl	Dyn	Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	2.54	2.58 (55)	1.98 (50)	**9.56**	**9.21** (52)	**5.40** (49)	**9.58**	**5.33** (50)	2.42 (54)
H_2SO_4	2	2.01	1.47 (48)	1.33 (49)	2.22	1.68 (51)	1.44 (44)	1.83	1.64 (51)	1.24 (50)
Na_2SO_4	1	**10.39**	**6.60** (50)	2.07 (45)	**11.00**	4.99 (49)	2.59 (48)	**9.93**	**5.72** (51)	2.26 (51)
Na_2SO_4	2	3.92	1.82 (49)	1.66 (47)	**9.51**	3.02 (48)	2.16 (50)	**7.61**	2.07 (48)	2.07 (53)
$ZnSO_4$	1	**95.61**	**69.59** (47)	**71.93** (52)	**41.05**	**46.68** (51)	**48.95** (52)	**41.23**	**40.31** (49)	**39.82** (52)
$ZnSO_4$	2	**89.82**	**82.00** (48)	**93.86** (58)	**60.35**	**73.55** (50)	**72.98** (50)	**40.53**	**38.49** (53)	**35.09** (55)

of the retrains. As the retrains must be made in average each 45–60 samples, we adopt the same range [*floor*(60/*c*), 60] as in Section 4.1 for the search, and we determine μ empirically.

- *Dynamic*: Our active learning strategy, described in detail in Section 4.1

6. Results

6.1. Predictive model quality

6.1.1. Incremental stage

Tables 2–4 provide, respectively for window sizes 50, 100 and 200, a summary of the normalized root mean squared errors (NRMSE) calculated over the whole data sets, based on the *k*-step ahead predictions and obtained residuals. These errors were calculated for all three outgoing sample selection strategies: *Random*, *Oldest*, and *LessInfo* (our approach). For each of them, all three ingoing sample selection methods were tested: *Static*, *Euclidean*, and *Dynamic* (our approach). Table 5 reports the same errors for the two non-window-based state-of-the-art incremental algorithms (*IncrPCR* and *eChemo*) corresponding to the three ingoing sample selection approaches. The values in bold font indicate errors which are above the limit of 5% as a requirement of the company.

It is clear that for H_2SO_4 and Na_2SO_4 the error reduction when using our incremental sample selection is significant in several cases when applying FLEXFIS + PLS, whereas for $ZnSO_4$ the modeling fails as achieving error rates of minimal 20% or higher, for WS = 50, and over 5% in the other cases. The explanation of this occurrence is the much more dynamic behavior of $ZnSO_4$, as can be seen in Fig. 7. Moreover, FLEXFIS needs bigger window sizes in order to construct clusters, thus rules, that are representing the data significantly. It is then clear that the FLEXFIS results, for WS = 200, are at least as good as the ones from PLSR, and sometimes better. Also, the performance of PLS is a bit improved with bigger windows. Also a big window size helps in the case of $ZnSO_4$, because the initial training set covers the range better. Nevertheless, it is not possible to reach the 5% required by the company, but at least coming close to it (around 6% and slightly below). On the other hand, we will see (see Section 6.2 below) that bigger window sizes lead to a slower computational performance. Our incremental sample selection concepts sometimes even act as "peak smoother" in the

Table 3
Normalized root mean squared errors (percentual errors) for PLSR (upper part) and FLEXFIS + PLS (lower part) in case of a window size of 100, using different variants for sample exclusion (oldest, random and our decremental active learning ("LessInfo")) and applying equidistant sample accumulation according to a re-train ratio of 60 ("Stat" columns), the approach in [21] ("Eucl" columns), and the exquisite selection according to our incremental active learning topology ("Dyn" columns). The values inside the parenthesis indicate the real averaged retrain ratio achieved (note that that value is fixed in 60 for "Stat"), and the values in bold font indicate violations of company requirement (max. 5% errors).

PLSR		Random			Oldest			LessInfo		
Target	Set	Stat	Eucl	Dyn	Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	1.31	1.32 (53)	1.31 (54)	1.32	1.29 (52)	1.30 (54)	1.30	1.32 (54)	1.32 (52)
H_2SO_4	2	1.15	1.18 (54)	1.17 (53)	1.16	1.16 (57)	1.16 (58)	1.07	1.05 (52)	1.00 (50)
Na_2SO_4	1	1.68	1.66 (47)	1.67 (51)	1.44	1.49 (57)	1.45 (57)	1.42	1.44 (58)	1.37 (53)
Na_2SO_4	2	1.67	1.62 (50)	1.60 (51)	1.65	1.67 (52)	1.61 (55)	1.62	1.58 (52)	1.59 (52)
$ZnSO_4$	1	**6.47**	**6.32** (52)	**5.95** (53)	**6.57**	**5.99** (47)	**6.02** (52)	**6.22**	**6.49** (53)	**5.97** (55)
$ZnSO_4$	2	**5.96**	**5.95** (53)	**5.94** (52)	**6.12**	**5.96** (53)	**5.99** (53)	**5.93**	**5.91** (49)	**5.89** (54)
FLEXFIS		Random			Oldest			LessInfo		
Target	Set	Stat	Eucl	Dyn	Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	1.45	1.34 (53)	1.32 (52)	1.51	1.54 (49)	1.35 (51)	1.50	1.45 (54)	1.47 (54)
H_2SO_4	2	1.30	1.15 (54)	1.08 (53)	1.99	2.01 (58)	1.47 (53)	1.63	1.48 (52)	1.21 (51)
Na_2SO_4	1	3.75	2.12 (52)	2.03 (50)	3.81	2.01 (51)	1.86 (50)	2.74	1.73 (52)	1.70 (53)
Na_2SO_4	2	2.24	1.69 (50)	1.62 (51)	3.92	3.17 (54)	2.10 (50)	2.66	1.49 (47)	1.58 (52)
$ZnSO_4$	1	**8.97**	**7.95** (51)	**7.88** (50)	**6.87**	**5.92** (53)	**5.98** (52)	**6.53**	**5.95** (48)	**5.95** (50)
$ZnSO_4$	2	**8.41**	**7.52** (54)	**7.20** (55)	**5.99**	**5.85** (52)	**5.89** (53)	**5.92**	**5.94** (54)	**5.86** (51)

Table 4
Normalized root mean squared errors (percentual errors) for PLSR (upper part) and FLEXFIS + PLS (lower part) in case of a window size of 200, using different variants for sample exclusion (oldest, random and our decremental active learning ("LessInfo")) and applying equidistant sample accumulation according to a re-train ratio of 60 ("Stat" columns), the approach in [21] ("Eucl" columns), and the exquisite selection according to our incremental active learning topology ("Dyn" columns). The values inside the parenthesis indicate the real averaged retrain ratio achieved (note that that value is fixed in 60 for "Stat"), and the values in bold font indicate violations of company requirement (max. 5% errors).

PLSR		Random			Oldest			LessInfo		
Target	Set	Stat	Eucl	Dyn	Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	1.32	1.29 (49)	1.30 (52)	1.30	1.29 (50)	1.28 (51)	1.27	1.21 (48)	1.21 (50)
H_2SO_4	2	1.13	1.15 (58)	1.10 (52)	1.09	1.06 (54)	1.04 (54)	1.03	0.98 (50)	0.92 (49)
Na_2SO_4	1	1.71	1.70 (50)	1.71 (52)	1.42	1.39 (54)	1.39 (53)	1.34	1.36 (58)	1.35 (54)
Na_2SO_4	2	1.66	1.66 (52)	1.66 (52)	1.48	1.53 (49)	1.50 (53)	1.60	1.54 (52)	1.49 (51)
$ZnSO_4$	1	**6.03**	**5.98** (51)	**5.90** (54)	**6.10**	**5.05** (53)	**6.01** (52)	**6.13**	**5.97** (51)	**5.99** (51)
$ZnSO_4$	2	**5.93**	**5.91** (53)	**5.85** (55)	**6.03**	**5.92** (53)	**5.92** (53)	**5.91**	**5.91** (52)	**5.89** (54)
FLEXFIS		Random			Oldest			LessInfo		
Target	Set	Stat	Eucl	Dyn	Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	1.37	1.34 (53)	1.30 (53)	1.29	1.20 (50)	1.25 (51)	1.26	1.21 (47)	1.19 (49)
H_2SO_4	2	1.19	1.17 (50)	1.06 (51)	1.07	1.08 (54)	1.00 (53)	1.05	1.00 (49)	0.99 (50)
Na_2SO_4	1	1.77	1.73 (54)	1.66 (53)	1.74	1.36 (49)	1.37 (53)	1.66	1.38 (50)	1.37 (51)
Na_2SO_4	2	1.66	1.65 (52)	1.59 (50)	3.01	2.22 (54)	2.00 (56)	1.91	1.64 (53)	1.52 (52)
$ZnSO_4$	1	**6.97**	**6.83** (53)	**6.28** (52)	**5.96**	**5.95** (51)	**5.94** (53)	**5.20**	**5.25** (53)	**5.03** (53)
$ZnSO_4$	2	**6.10**	**5.85** (52)	**5.87** (54)	**5.89**	**5.88** (51)	**5.77** (53)	**6.00**	**5.77** (53)	**5.77** (51)

residual space, one example plot is shown in Fig. 8, where peaks in the predictions (due to distinct extrapolation cases) can be completed avoided. Here, also the correlation between the observed and predicted values is significantly improved, as the predicted ones follow much better the trend of the real observed (measured) target values. The Euclidean ingoing sample selection usually performs better than the static (removing also most of the peaks), but most of the times it performs slightly worse than dynamic, even when using Euclidean implies almost always a lower average retrain ratio (the numbers inside the parenthesis) thus more total number of retrains during the whole process. This is especially the case when using our new approach "LessInfo" (proposed in Section 4.2) in combination with a small window size of 50. This combination variant is the most beneficial one, as 1.) we will see in the subsequent section that "LessInfo" avoids big prediction errors which is not the case for the other approaches and 2.) a window size of 50 achieves computation times which are in the same range of the fast incremental approaches (*eChemo* and incrPCR), especially when PLS is used as modeling variant (see Section 6.2).

6.1.2. Decremental stage

From first glance it seems that the more intelligence regarding sample exclusion brought in by our decremental active learning concepts (termed as "LessInfo") does not pay-off in lower errors, neither for linear models (PLS) nor for non-linear ones (FLEXFIS + PLS). We verified this statistically on a sample-wise level (using each residual as one sample in the statistical test) employing Mann–Whitney–Wilcoxon (M–W–W) test [46], which confirms this observation of no preference at a significance level of $\alpha = 0.01$. However, when inspecting the distribution of the errors in more detail, we found out that more medium to large errors are caused by random and oldest selection than with our approach. A visualization of error histograms is presented in Fig. 9 for H_2SO_4 and in Fig. 10 for Na_2SO_4. Finally, IncrPCR and *eChemo* do not achieve any meaningful error rates, regardless of the ingoing sample selection employed, because they would need more frequent adaptations in order to improve their accuracy—see Table 5 (very high errors).

Table 5
Normalized root mean squared errors (percentual errors) for IncrPCR and *eChemo*, and for all three ingoing sample selection approaches are reported: applying equidistant sample accumulation according to a re-train ratio of 60 ("Stat" columns), the approach in [21] ("Eucl" columns), and the exquisite selection according to our incremental active learning topology ("Dyn" columns). Notice that these methods are no window-based, thus there is not outgoing sample selection running, but they precise an initial batch model (made with the first 200 samples). The values inside the parenthesis indicate the real averaged update ratio achieved (note that that value is fixed in 60 for "Stat"), and the values in bold font indicate violations of company requirement (max. 5% errors).

Target	Set	IncrPCR			*eChemo*		
		Stat	Eucl	Dyn	Stat	Eucl	Dyn
H_2SO_4	1	**32.12**	**30.69** (52)	**30.37** (53)	**30.33**	**28.47** (50)	**27.99** (51)
H_2SO_4	1	**31.77**	**32.02** (55)	**31.83** (54)	**29.62**	**26.52** (51)	**24.61** (53)
Na_2SO_4	1	**49.56**	**49.10** (51)	**48.53** (52)	**47.24**	**46.78** (53)	**44.25** (52)
Na_2SO_4	1	**48.41**	**48.37** (55)	**46.43** (55)	**46.15**	**42.57** (50)	**39.14** (53)
$ZnSO_4$	1	**77.64**	**75.20** (53)	**73.48** (54)	**69.78**	**68.37** (52)	**62.79** (51)
$ZnSO_4$	1	**77.14**	**71.56** (52)	**71.62** (52)	**68.44**	**68.01** (53)	**62.55** (49)

6.2. Computation speed

Since we are monitoring an on-line process, computation times are crucial. Table 6 shows the averaged computation times per retrain, for equidistant static sample accumulation with a retrain ratio of 60 (left), Euclidean ingoing sample selection (center), and our incremental dynamic active learning approach (right). These values are provided for each of the algorithms and targets. The upper part of the table contains values for the window-based approaches (for all three WS values, and our decremental active learning approach because random and oldest outgoing sample selection methods are faster even), and the lower part for the incremental non-window-based methods with a BS of 200. Notice that the time for the predictions of the samples previous to the incoming sample is also included. As there are no significant differences, in terms of computational times, between Set 1 and Set 2, because they have the same number of variables in every target (as reported in Table 1), the values showed correspond to the average between Set 1 and Set 2 for each target. All the tests were run under MATLAB, using a 64-bit Intel ®i7 QuadCore 2.8 GHz PC.

It is clear, as could be obviously expected, that the incremental approaches are faster than the ones based on the sliding window, especially in the case of FLEXFIS, because the latter ones are retraining based strategies. But since we have to fulfill some accuracy requirements, the former are not useful, compare with Tables 2 to 4. Adapting more frequently, as was proved in [2], we could reach those accuracy requirements, achieving similar results as with the sliding window concept, but the costs would be unaffordable. The reason for this big difference in performance lies in the less flexibility of *eChemo*, as no dynamic input structure change and no adaptive learning parameter (steering trade-off between rule evolution and model adaptation) are integrated—in

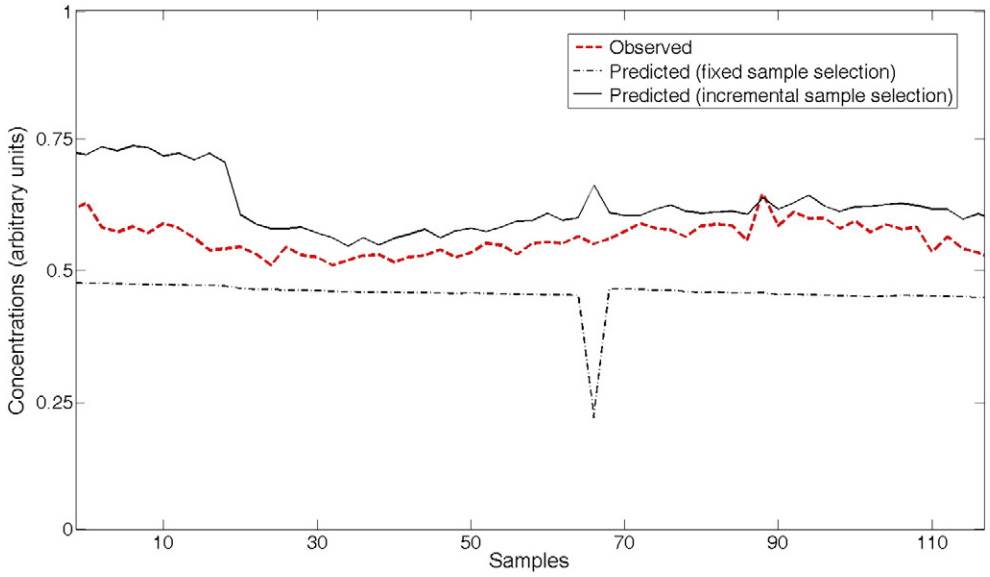


Fig. 8. Peak deletion example for Na_2SO_4 when using incremental (solid) instead of pre-fixed sample selection (dash–dot), also a achieving a higher correlation between observed and predicted values.

our new variant, both are optimized within each re-training cycle phase according to best-parameter-grid-search scenario.

Since in the current installation our time margin is around 8 min, we are perfectly on time for applying any of the sliding window based strategies, i.e. a re-calibration cycle is surely finished even in the worst case when the next one is triggered. Moreover, if this time margin was reduced in the future, we could also speed up our processes by reducing the grid in the grid search used for the best parameters selection because those optimum parameters are not very dynamic in consecutive retrains and thus can be transferred to the subsequent ones.

7. Conclusion

In this paper, we are proposing a sliding window based approach which comes with decremental and incremental active learning stages for ideally selecting those samples for exclusion and inclusion into the

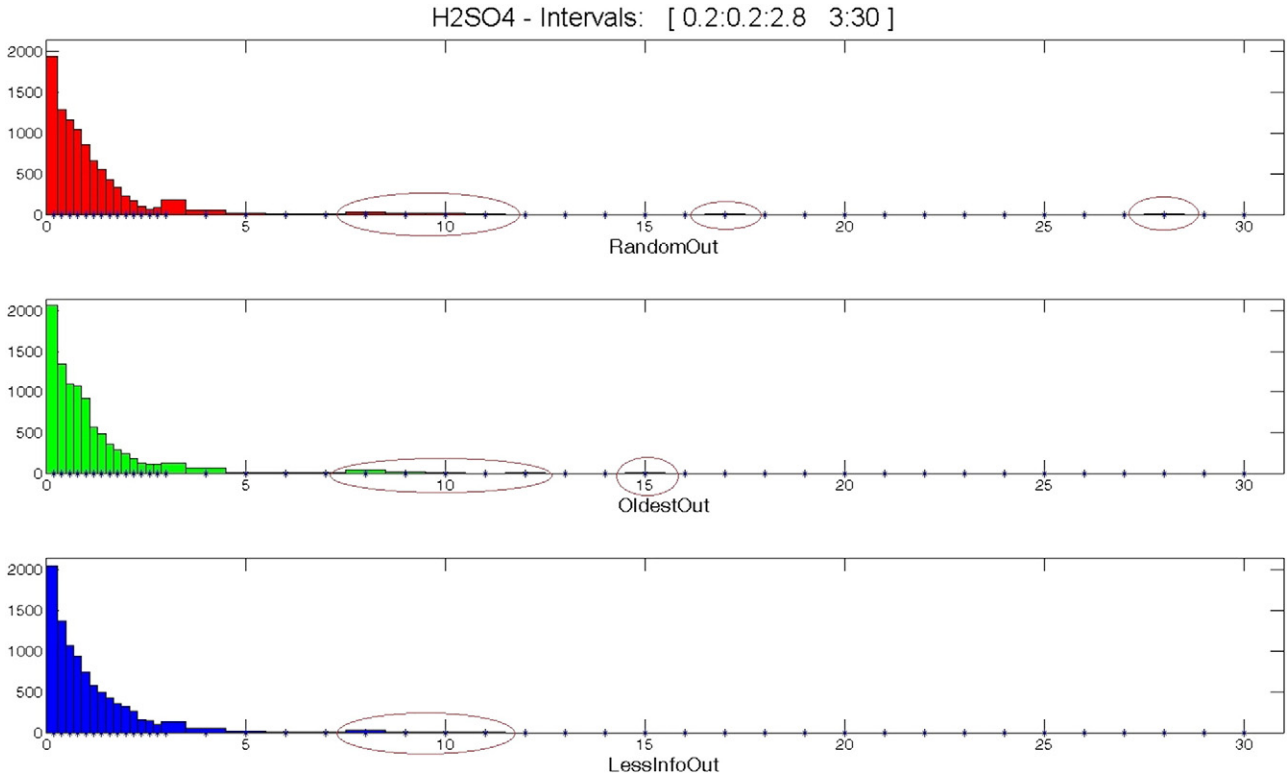


Fig. 9. Error histograms (based on NRMSE values) for H_2SO_4 using the three different sample exclusion methods, ellipses indicting high errors.

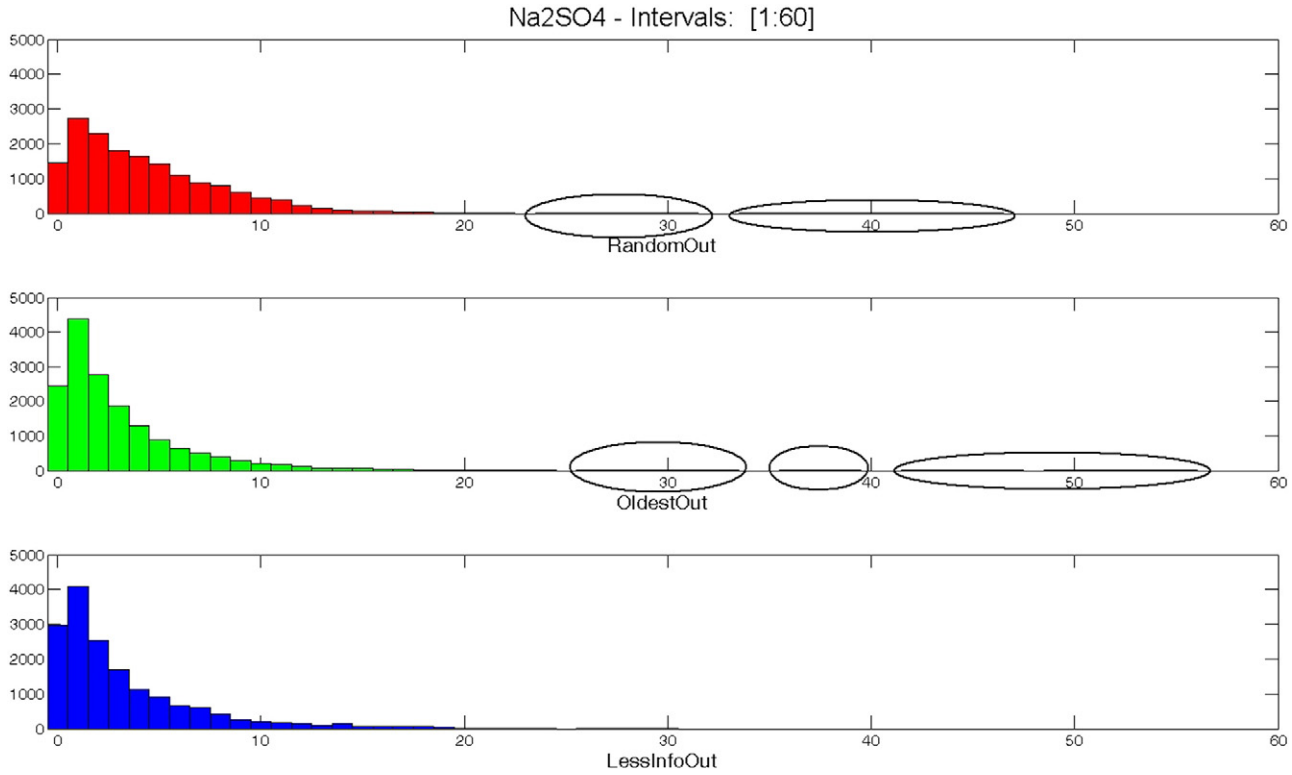


Fig. 10. Error histograms (based on NRMSE values) for Na_2SO_4 using the three different sample exclusion methods, ellipses indicting high errors.

sliding window. The selections are based on minimizing information loss for the model trained from the samples contained in the actual window (decremental stage), respectively on optimizing information gain (incremental stage). The incremental stage is restricted by a minimal allowed inclusion frequency according to company requests (in average 3 to 4 times a day). This leads to a significant cost reduction in terms of saving measurements produced by the automated titration, while still achieving error rates far below the requested 5% for H_2SO_4 and Na_2SO_4 (for $ZnSO_4$ they are slightly above, however). In terms of money, this means that the savings are expected by about more than 100 K Euro per year, which cannot be achieved with state-of-the-art methods (very high errors are then caused).

Future work includes the development of a complete unsupervised adaptation approach, which together with an initial supervised stage (training from first d samples), leads to a semi-supervised approach. This variant may then completely abandon the usage of any target values, making the automated titration superfluous and reducing further the company costs, especially in terms of equipment acquisition.

Conflict of interest

There no conflicts of interest for this paper.

Acknowledgments

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Table 6
Averaged computation times (in seconds), for each window size, for sliding window based approaches (upper part), and for incremental, evolving variants IncrPCR and *eChemo*, for initial batch size $BS = 200$ (lower part). Each cell shows values corresponding to all three ingoing sample selection procedures, i.e. Static/Euclidean/Dynamic.

WS	PLSR (H_2SO_4)	FLEX (H_2SO_4)	PLSR (Na_2SO_4)	FLEX (Na_2SO_4)	PLSR ($ZnSO_4$)	FLEX ($ZnSO_4$)
50	2.96/3.38/3.57	26.75/29.58/31.40	3.59/4.26/4.42	33.12/37.91/39.52	3.08/3.79/3.87	28.31/32.78/34.32
100	6.68/7.28/7.80	55.27/67.24/68.46	7.82/9.01/9.57	71.93/77.96/79.41	6.80/7.83/8.28	59.92/73.33/74.84
200	15.99/18.02/18.84	126.62/147.21/148.22	19.13/22.90/23.71	158.94/170.27/171.36	16.88/20.69/21.27	131.11/160.86/162.45
BS	IncrPCR (H_2SO_4)	*eChemo* (H_2SO_4)	IncrPCR (Na_2SO_4)	*eChemo* (Na_2SO_4)	IncrPCR ($ZnSO_4$)	*eChemo* ($ZnSO_4$)
200	3.81/4.37/4.52	0.93/1.42/1.55	3.98/4.48/4.59	1.05/1.54/1.61	3.73/4.35/4.48	0.97/1.45/1.56

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